

China-ASEAN Advanced Course on Ocean Law and Management



Maritime Administration & International Maritime Regulations of IMO

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Dalian Maritime University

NISCSS, Haikou, China, 27 Jan 2016



International Maritime Regulations of IMO



1. International Maritime Organization
2. IMO Structure
3. IMO Key Conventions
4. Current Hot Issues

1. International Maritime Organization (IMO)



International Maritime Regulations of IMO

1948

- UN adopted Convention for **IMCO**

1949

- IMCO first meeting

1982

- Renamed **IMO**





International Maritime Regulations of IMO



IMO is the UN's specialized agency with responsibility for the safety and security of shipping and the prevention of marine pollution by ships.





IMO's main role is to create a regulatory framework for the shipping industry, which is universally adopted and implemented.

Mission Statement:

to promote safe,
secure,
environmentally sound,
efficient and
sustainable shipping
through cooperation.



International Maritime Regulations of IMO



171 Member States



3 Associate Members



65 IGOs



77 NGOs

Slogan:

Safe, Secure and Efficient Shipping on Clean Oceans.

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IMO WORLD MARITIME DAY 2016

SHIPPING

INDISPENSABLE TO THE WORLD

Theme of

World Maritime Day 2016

International Maritime Regulations of IMO

Former Secretary-Generals of IMO

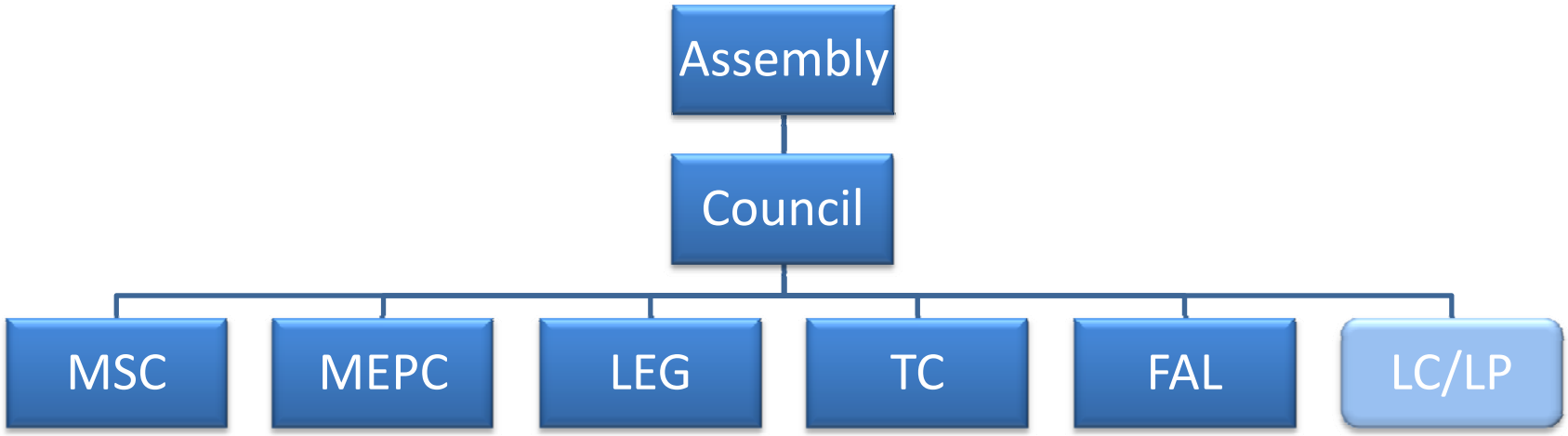




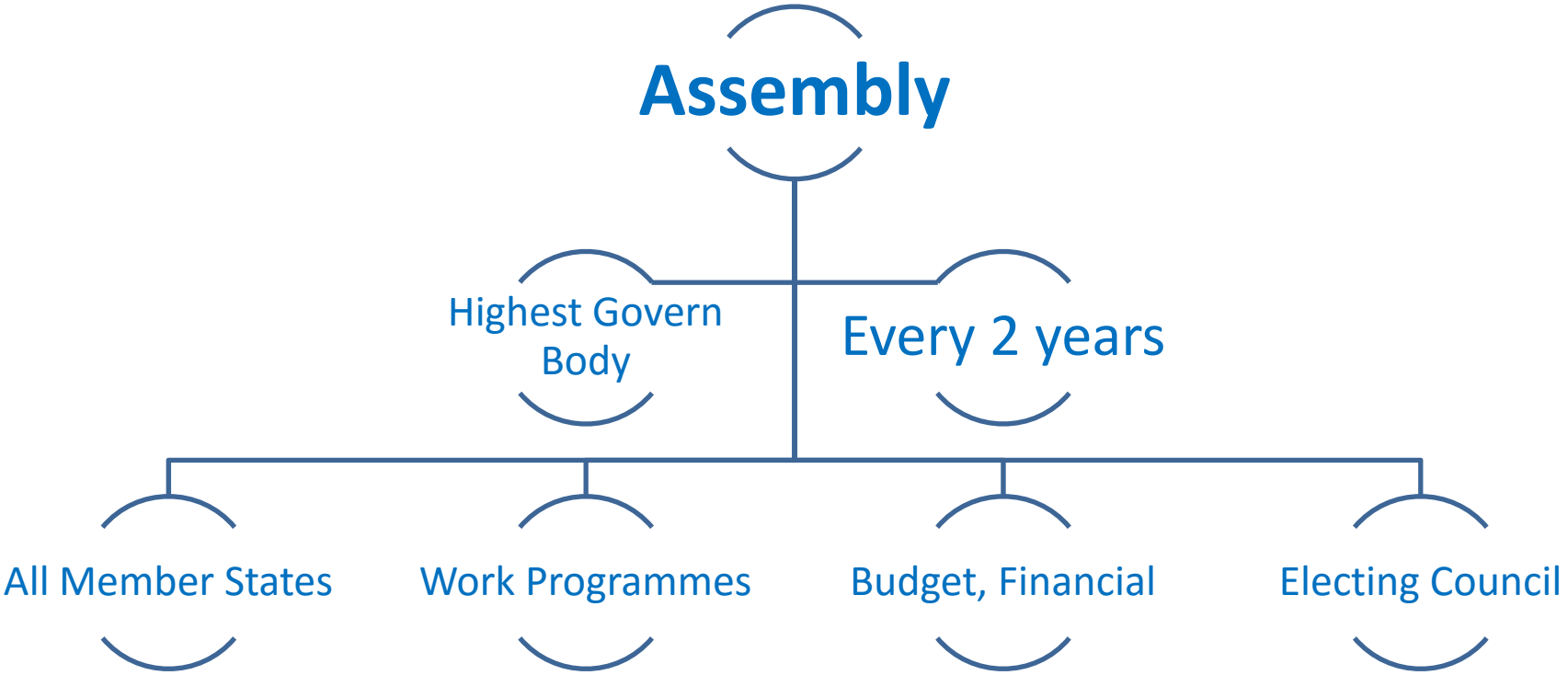
2. IMO Structure



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Council Members for 2016-2017 Biennium

Council Category

A: 10, China, Japan, Korea …

B: 10, Bangladesh, India …

C: 20, Indonesia, Malaysia, Philippines, Singapore, Thailand …

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MEPC



Oil Spill



BWM

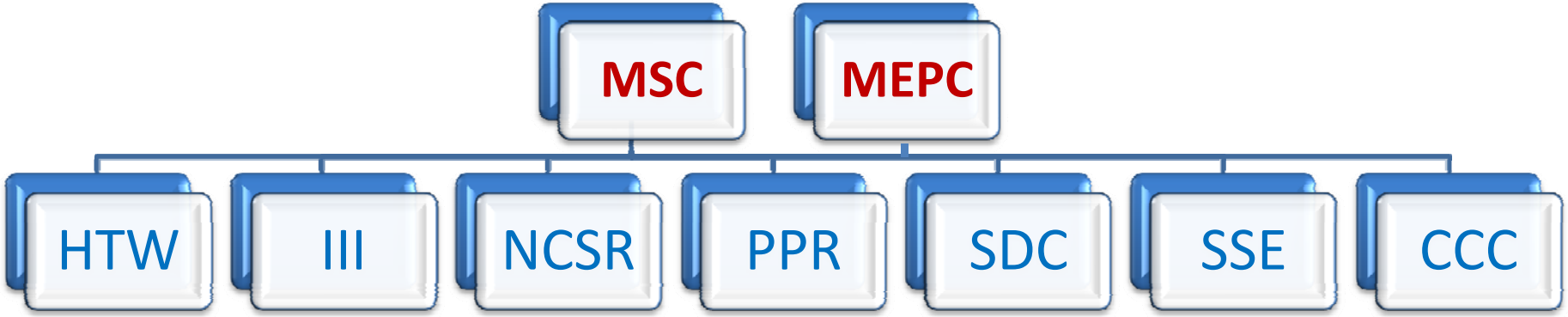


Polar Code



EEDI

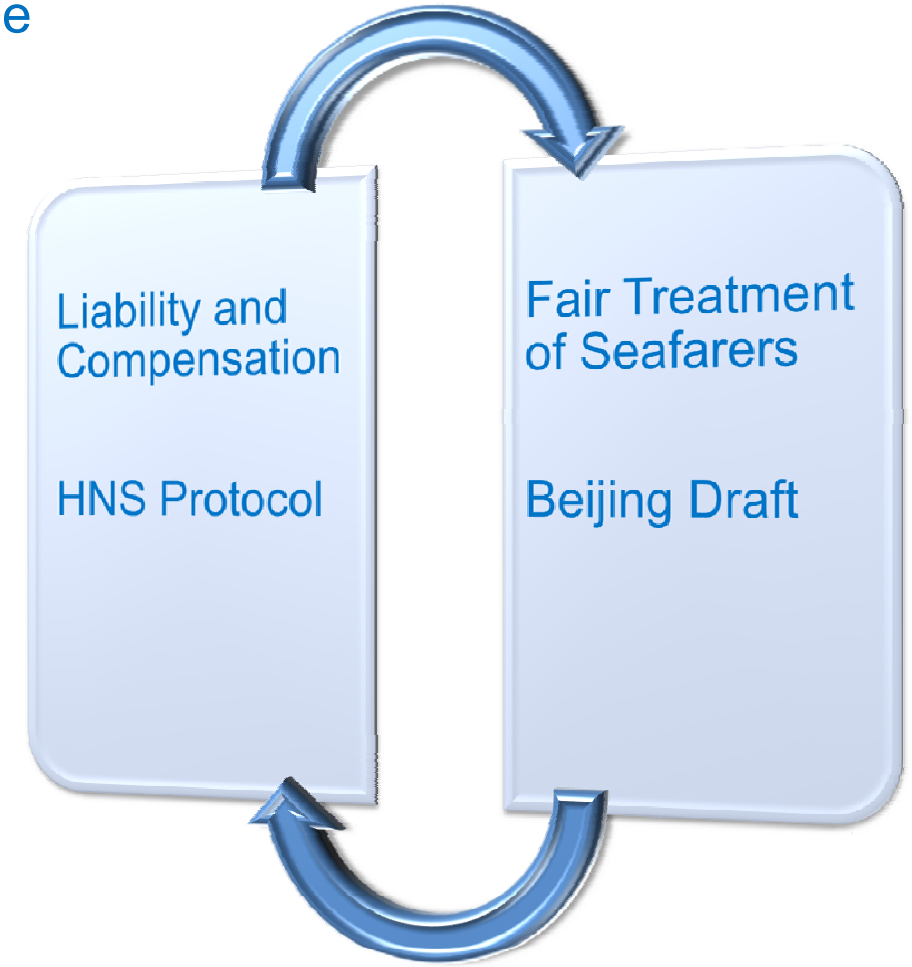
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HTW – Human element, Training and Watchkeeping
III – Implementation of IMO's Instruments
NCSR - Navigation, Communications and Search and Rescue
PPR - Pollution Prevention and Response
SDC - Ship Design and Construction
SSE - Ship Systems and Equipment
CCC - Carriage of Cargoes and Containers

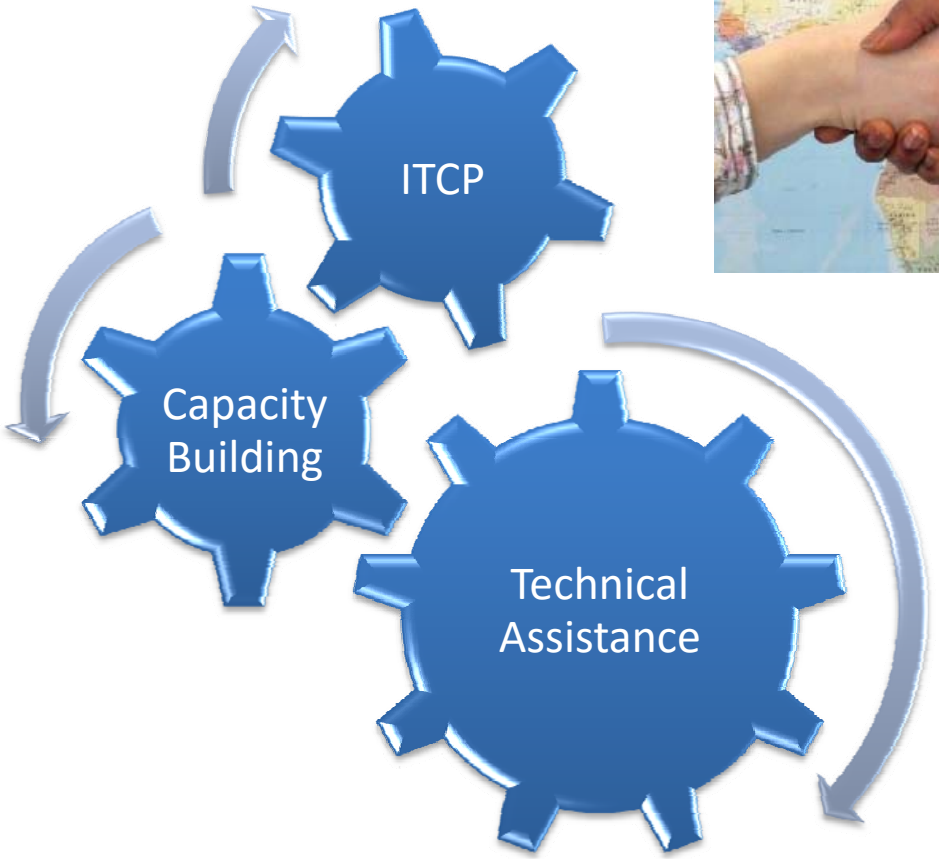
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Legal Committee



Technical Cooperation Committee

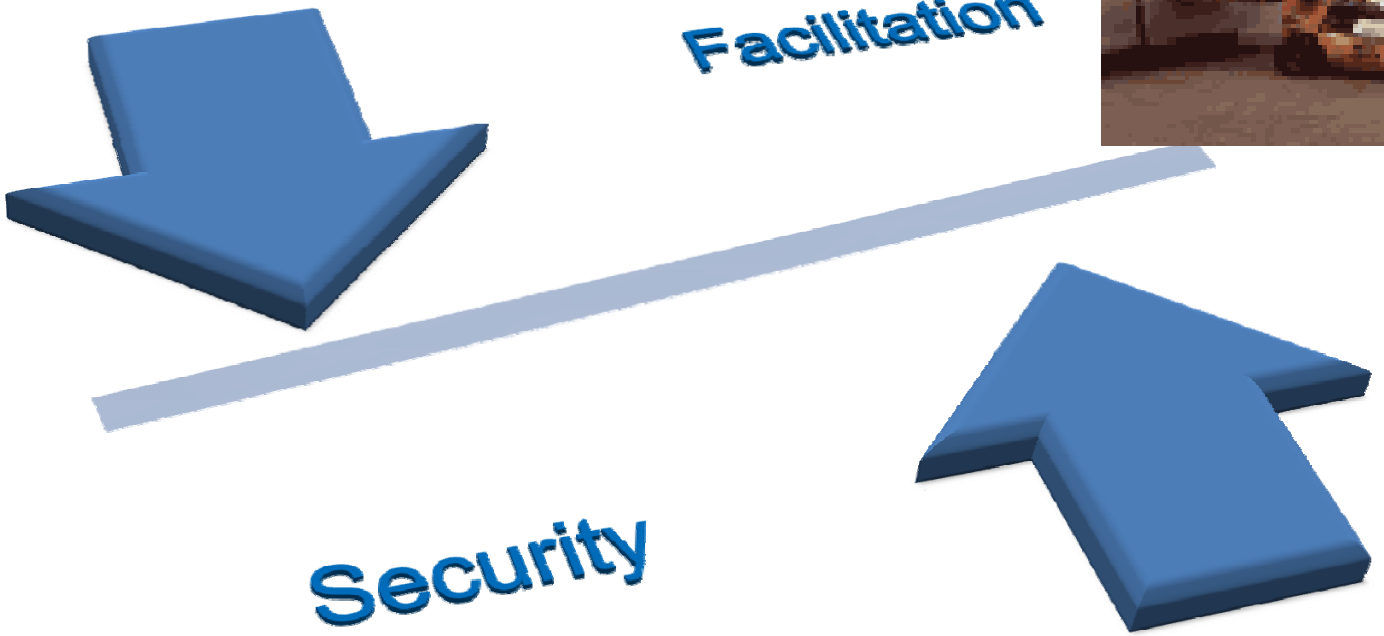




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Facilitation Committee







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S-G

Mr. Ki-Tack Lim,
New S-G of IMO
1 Jan 2016 -

S-G OFF

Audit OFF

MSD

MED

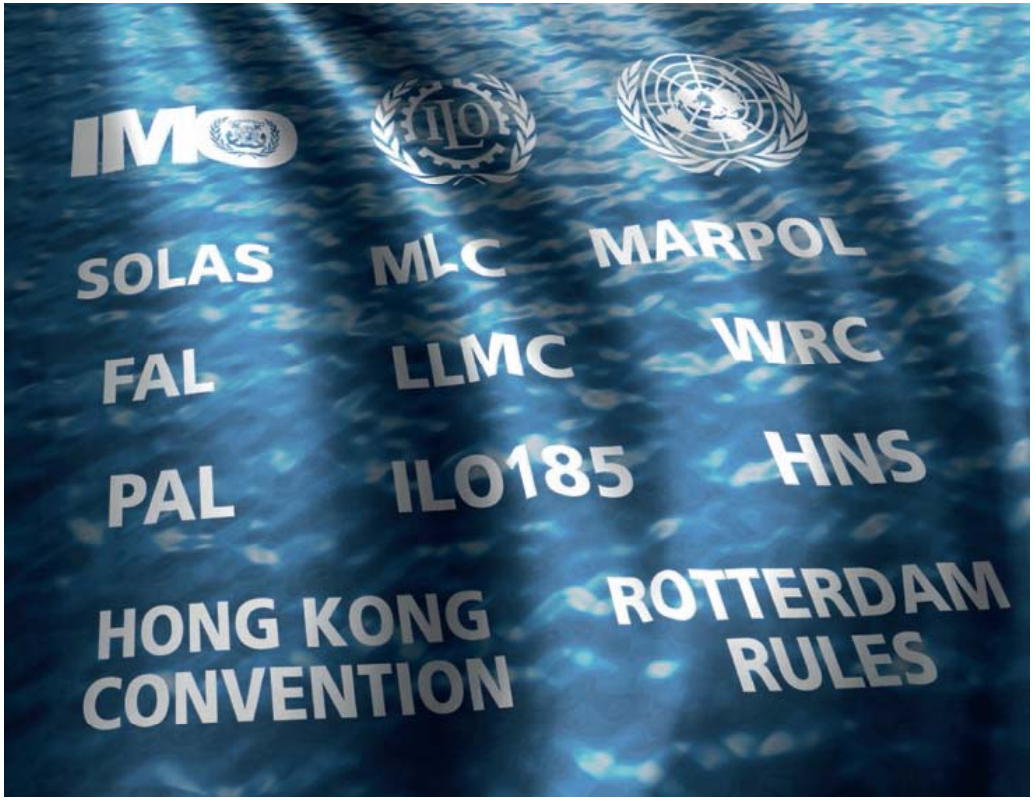
E&LD

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HR

3. IMO Conventions



IMO Conventions:

- **59 Conventions / Protocols in total**
  - 49 of them are in force
  - 7 of them are NOT yet in force
  - 3 of them are NOT intended to enter into force

IMO plays depositary function for London Convention/Protocol.
IMO is a joint depositary with ICAO for Cospas-Sarsat Agreement.



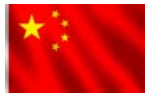


IMO Conventions are grouped as follows:

- Key IMO Conventions
- Conventions relating to Maritime Safety and Security
- Conventions relating to Prevention of Marine Pollution
- Conventions covering Liability and Compensation
- Other Conventions

Key IMO Conventions (3)

- The International Convention for the Safety of Life at Sea (SOLAS), as amended
- The International Convention for the Prevention of Pollution from Ships 1973, as modified by the 1978 & 1997 Protocols (MARPOL),
- The International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW), 1978, as amended in 1995 and 2010 (Manila Amendments)





Conventions relating to Maritime Safety and Security (10)

- Convention on the International Regulations for Preventing Collisions at Sea (COLREG), 1972
- Convention on Facilitation of International Maritime Traffic (FAL), 1965
- International Convention on Load Lines (LL), 1966
- International Convention on Maritime Search and Rescue (SAR), 1979
- Convention for the Suppression of Unlawful Acts Against the Safety of Maritime Navigation (SUA), 1988, and Protocol for the Suppression of Unlawful Acts Against the Safety of Fixed Platforms located on the Continental Shelf (and the 2005 Protocols)
- International Convention for Safe Containers (CSC), 1972

Conventions relating to Maritime Safety and Security: (Continued)

- Convention on the International Maritime Satellite Organization (IMSO C), 1976
- The Torremolinos International Convention for the Safety of Fishing Vessels (SFV), 1977, superseded by the 1993 Torremolinos Protocol; Cape Town Agreement of 2012 on the Implementation of the Provisions of the 1993 Protocol relating to the Torremolinos International Convention for the Safety of Fishing Vessels
- International Convention on Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel (STCW-F), 1995
- Special Trade Passenger Ships Agreement (STP), 1971 and Protocol on Space Requirements for Special Trade Passenger Ships, 1973

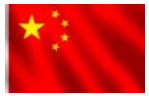


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Conventions relating to Prevention of Marine Pollution (7)

- International Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties (INTERVENTION) 1969
- Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter (LC), 1972 (and the 1996 London Protocol)
- International Convention on Oil Pollution Preparedness, Response and Co-operation (OPRC), 1990
- Protocol on Preparedness, Response and Co-operation to pollution Incidents by Hazardous and Noxious Substances, 2000 (OPRC-HNS Protocol)
- International Convention on the Control of Harmful Anti-fouling System on Ships (AFS), 2001
- International Convention for the Control and Management of Ship's Ballast Water and Sediments, 2004
- The Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, 2009



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Conventions covering Liability and Compensation (8)

- International Convention on Civil Liability for Oil Pollution Damage (CLC), 1969
- 1992 Protocol to the International Convention on the Establishment of an International Fund for Compensation for Oil Pollution Damage (FUND 1992)
- Convention relating to Civil Liability in the Field of Maritime Carriage of Nuclear Material (NUCLEAR), 1971
- Athens Convention relating to the Carriage of Passengers and their Luggage by Sea (PAL), 1974
- Convention on Limitation of Liability for Maritime Claims (LLMC), 1976
- International Convention on Liability and Compensation for Damage in Connection with the Carriage of Hazardous and Noxious Substances by Sea (HNS), 1996 (and its 2010 Protocol)
- International Convention on Civil Liability for Bunker Oil Pollution Damage, 2001
- Nairobi International Convention on the Removal of Wrecks, 2007

Other Conventions (2)

- International Convention on Tonnage Measurement of Ships (TONNAGE), 1969
- International Convention on Salvage (SALVAGE), 1989



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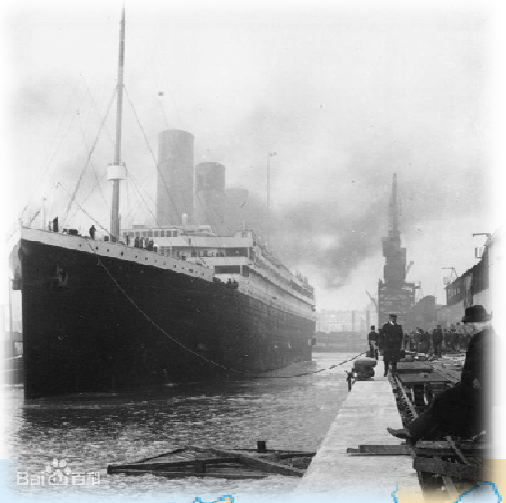


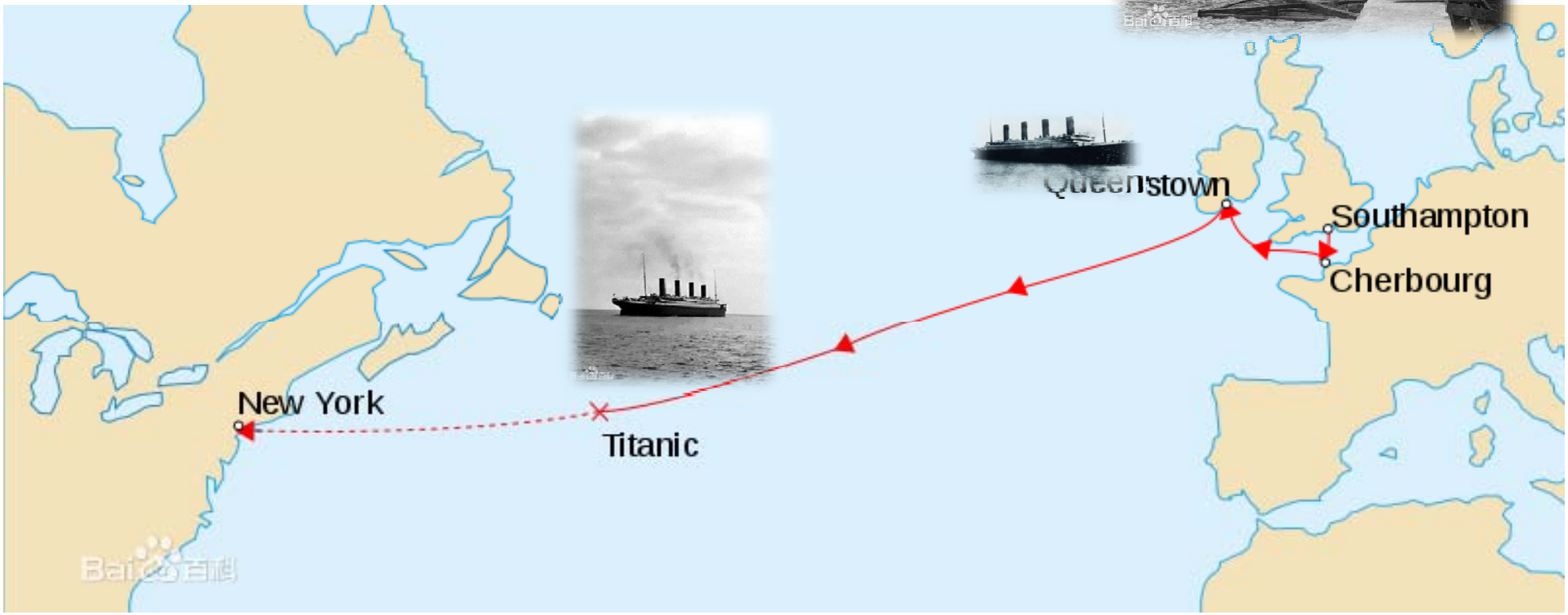
SOLAS is the most important international treaty concerning the safety of merchant ships.



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April 1912, Maiden Voyage,
15 April 1912, RMS Titanic sank
2207 passengers and crew
1517 people lost lives





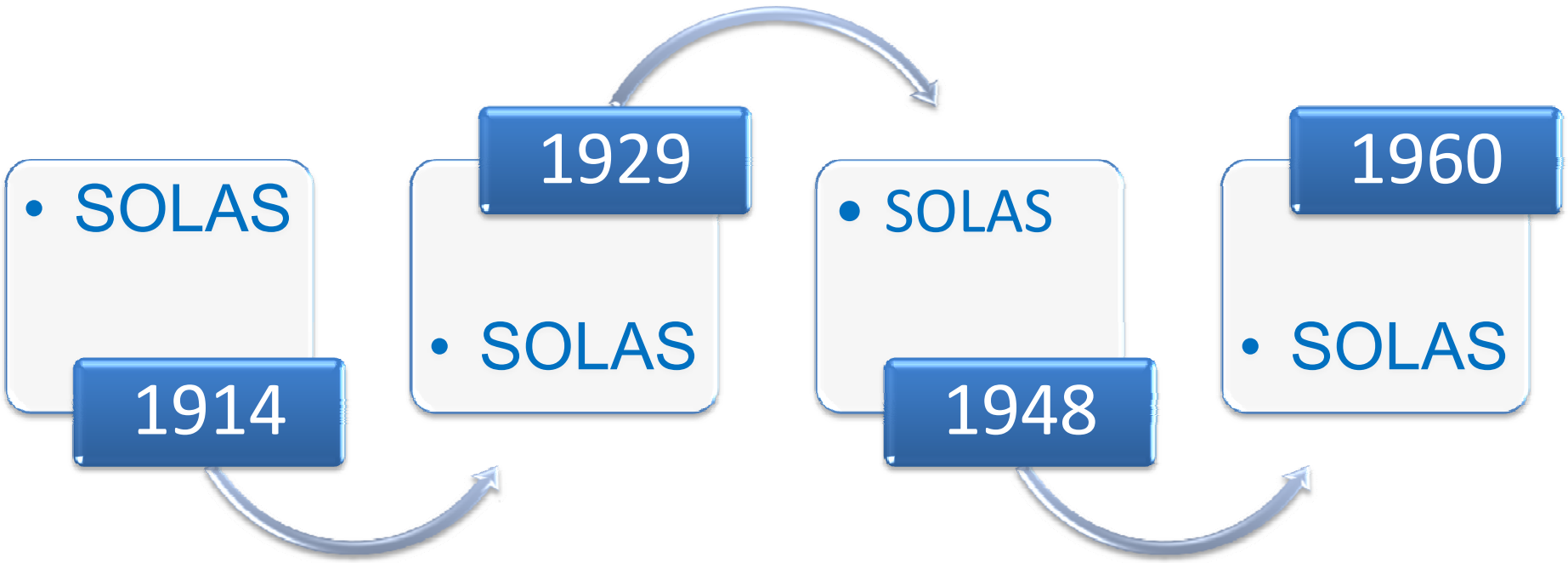




- The first version of SOLAS was adopted in 1914, in response to the Titanic disaster, the second in 1929, the third in 1948, and the fourth in 1960.
- A completely new Convention was adopted in 1974, with a new amendment procedure - the tacit acceptance procedure,
- The Convention in force today is sometimes referred to as SOLAS, 1974, as amended.

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SOLAS 1974 | Tacit acceptance



158 Amendments to SOLAS 1974



AMOCO CADIZ, 1978



Herald of Free Enterprise, 1987



Estonia, 1994



Scandinavian Star, 1990



International Maritime Regulations of IMO



Chapter I

- Chapter II
- Chapter III
- Chapter IV
- Chapter V
- Chapter VI

Chapter VII

- Chapter VIII
- Chapter IX
- Chapter X
- Chapter XI
- Chapter XII

New

- Chapter XIII
- Chapter XIV

Chapter I - General Provisions

- regulations concerning the survey of the various types of ships and
- issuing of documents signifying that the ship meets the requirements of the Convention.
- provisions for the control of ships in ports of other Contracting Governments.

Chapter II-1 Construction - Subdivision and stability, machinery and electrical installations

- subdivision of passenger ships into watertight compartments
- watertight integrity
- stability
- machinery & electrical
- steering gear

Chapter II-2 - Fire protection, fire detection and fire extinction

- fire safety measures
- division of the ship
- separation of accommodation
- restricted use of combustible materials;
- escape or access for fire-fighting purposes;
- fire-extinguishing appliances;

Chapter III - Life-saving appliances/arrangements

- requirements for life-saving appliances and arrangements,
- including life boats, rescue boats and life jackets according to type of ship.

Chapter IV – Radiocommunications

- Global Maritime Distress and Safety System (GMDSS).
- satellite emergency position indicating radio beacons (EPIRBs)
- search and rescue transponders (SARTs).

Chapter V - Safety of navigation

- navigation safety
- meteorological services for ships;
- ice patrol service;
- routeing of ships;
- search and rescue services.

Chapter VI - Carriage of cargoes

- all types of cargo
- stowage and securing cargo
- ships carrying grain to comply with the International Grain Code.

Chapter VII - Carriage of dangerous goods

- Part A - dangerous goods in packaged form
- Part A-1 - dangerous goods in solid form in bulk
- Part B - construction of ships carrying dangerous liquid chemicals in bulk and requires chemical tankers to comply with IBC Code
- Part C - construction of ships carrying liquefied gases in bulk and gas carriers to comply with IGC Code
- Part D - carriage of packaged irradiated nuclear fuel, radioactive wastes on board ships

Chapter VIII - Nuclear ships

- requirements for nuclear-powered ships
- particularly concern with radiation hazards.

Chapter IX - Management for the Safe Operation of Ships

- mandatory International Safety Management (ISM) Code,
- a safety management system to be established by the shipowner assuming responsibility for the ship

Chapter X - Safety measures for high-speed craft

- mandatory International Code of Safety for High-Speed Craft (HSC Code).

Chapter XI-1 - Special measures to enhance maritime safety

- authorization of recognized organizations (RO) , for surveys and inspections on Administrations' behalves;
- ship identification number scheme;
- port State control on operational requirements.



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Chapter XI-2 - Special measures to enhance maritime security

- adopted in December 2002, entered into force on 1 July 2004.
- International Ship and Port Facilities Security Code (ISPS Code).
- Part A of the Code is mandatory
- part B contains guidance.

Chapter XII - Additional safety measures for bulk carriers

- structural requirements for bulk carriers over 150 metres in length.

Chapter XIII – Verification of Compliance

- Implementation of IMO instruments
- IMO mandatory audit scheme

MSC.366(93) Resolution was adopted on 22 May 2014

The New Chapter XIII enters into force on 1 January 2016

Chapter XIV - Safety Measures for Ships Operating in Polar Waters

- Safety measures for ships in polar waters
- Polar Code: Part I (Safety)
  - A - safety measures
  - B - additional guidance

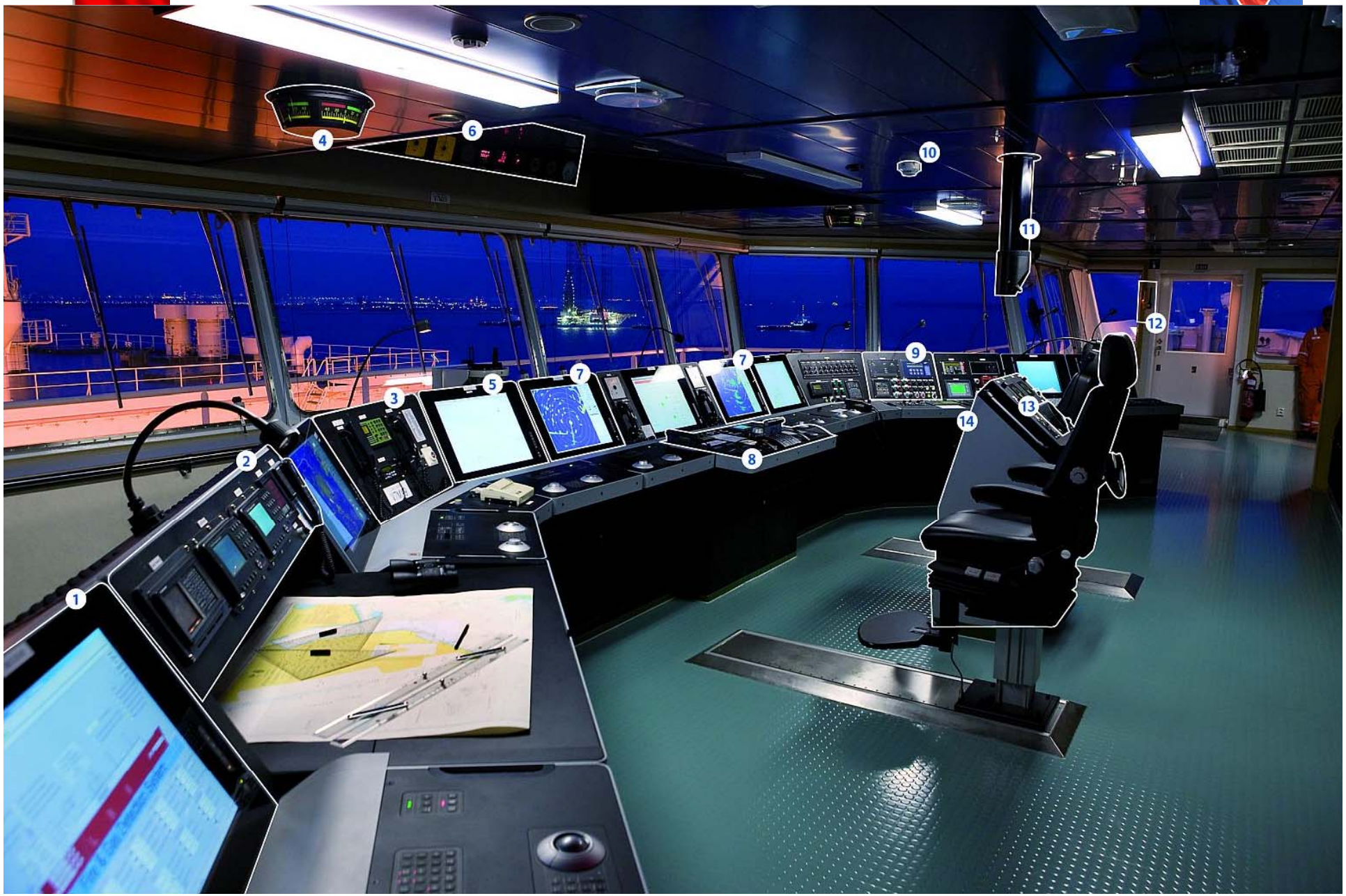


MSC 385/386 (94) adopted in Nov. 2014

MEPC 264/265 (68) adopted in May 2015,

Polar Code will enter into force on 1 Jan 2017

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Timeline: Key milestones in maritime safety since 1912

1914
International Convention for the Safety of Life at Sea (SOLAS) established – setting standards for maritime safety provisions.



1914
International Ice Patrol starts aerial monitoring of icebergs.

1922
Echo sounding applied on board to monitor depth of water.



1930
International Convention on Load Lines addresses issues on loading and stability.



1940s
Welding starts to replace riveting, later followed by prefabrication, increasing quality of ship construction.

1940s
LORAN (LOng RAnge Navigation) radio navigation system allows accurate offshore position finding to 900 miles.



Image: Wikimedia Commons/Stahlkocher

1944
DECCA position fixing allows accurate position finding up to 400 miles offshore.

1948
International Maritime Organization (IMO) established, and entered into force in 1958.



1960s
Computer-aided ship design revolutionizes ship design.

1960s
Widespread use of Very High Frequency radio improves ship-to-ship and ship-to-shore communication.

1965
RADAR made mandatory under 1960 SOLAS convention.

1967
"Transit" Sat Nav system: the first satellite-based positioning system for merchant ships, giving regular position fixes on 'transit' of a satellite.



1969
Automatic Radar Plotting Aid (ARPA) introduced (mandatory 1989), replacing manual plotting of movements.

1972
International Regulations for Preventing Collisions at Sea (COLREGS) establishes 'rules of the road' for shipping.

1973
International Convention for the Prevention of Pollution From Ships (MARPOL) addresses maritime pollution risk.



1978
International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW) establishes basic training and certification requirements.

1993
International Safety Management (ISM code) adopted by IMO, establishing standards for safe management and operation of ships.

1994
Global Positioning System (GPS) fully operational, allowing accurate satellite-based position finding.

1999
Global Maritime Distress and Safety System (GMDSS) establishes protocols for ships in distress and rescue scenarios and introduces mandatory distress communication equipment on board vessels.



Photo: Hervé Cozanet from the marine-marchande.net Website

2000
IMO adopts amendments to SOLAS making "Voyage Data Recorders" (VDR) or the 'Black Box' of navigational bridge mandatory on new ships.

2004
Automatic Identification System (AIS) for vessel identification and tracking, reduces collision risk.

2004
International Ship and Port Facility Security (ISPS) Code enhances security in ports.

2012
Electronic Chart Display and Information System (ECDIS) navigation system to become mandatory, providing continuous position and navigational information.

1910 1915 1920 1925 1930 1935 1940 1945 1950 1955 1960 1965 1970 1975 1980 1985 1990 1995 2000 2005 2010 2015

World Maritime Day 2012
100 Years after the Titanic

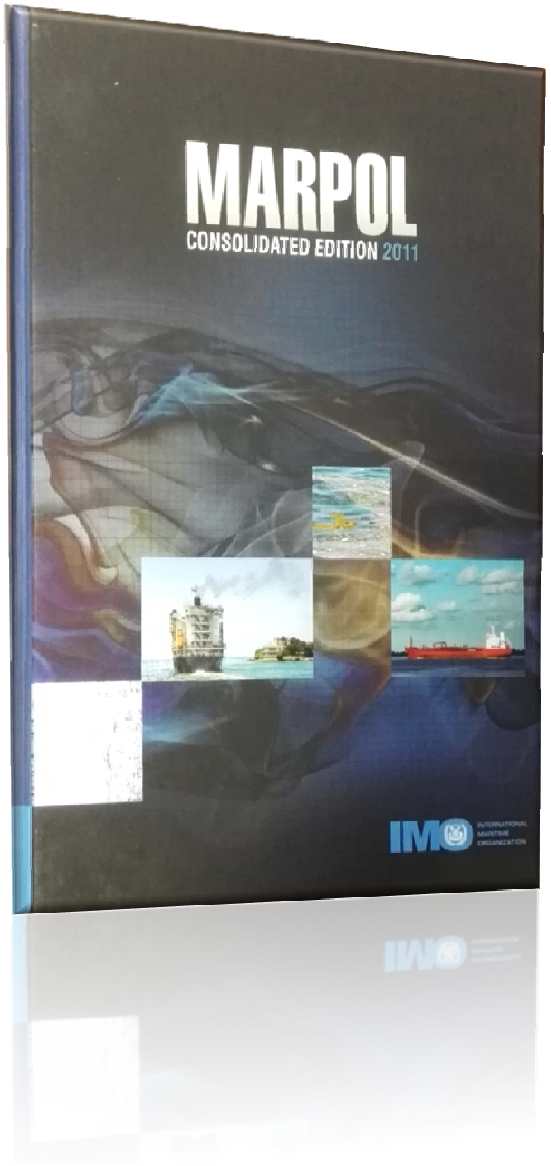




15 April 1912, Titanic

13 Jan 2012, Costa Concordia

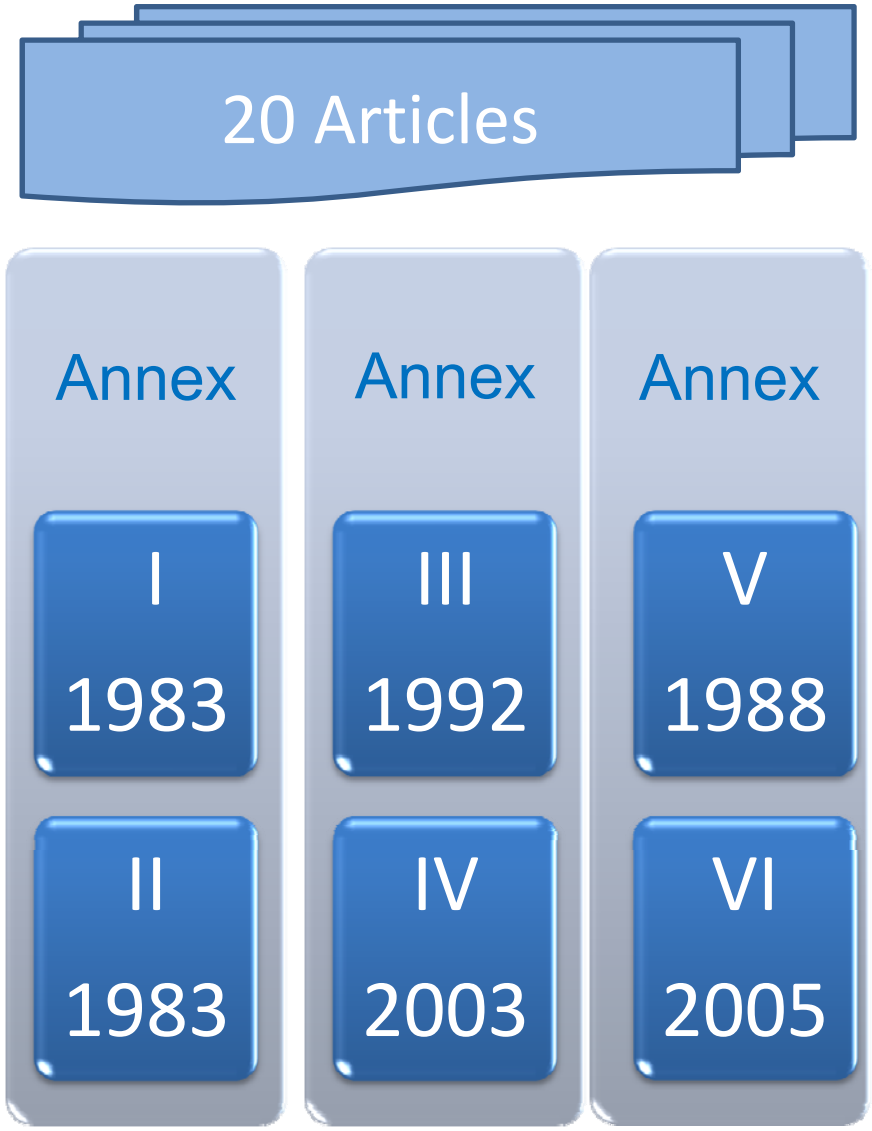
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MARPOL is the main international convention covering prevention of pollution of the marine environment by ships from operational or accidental causes.

MARPOL 73/78:

- Entry into force 2 Oct. 1983 (Annexes I & II).
- Annex III, entered into force 1 Jul. 1992,
- Annex IV, entered into force 27 Sept. 2003,
- Annex V, entered into force 31 Dec. 1988
- Annex VI, entered into force on 19 May 2005.



Annex I

Regulations for the Prevention of Pollution by Oil

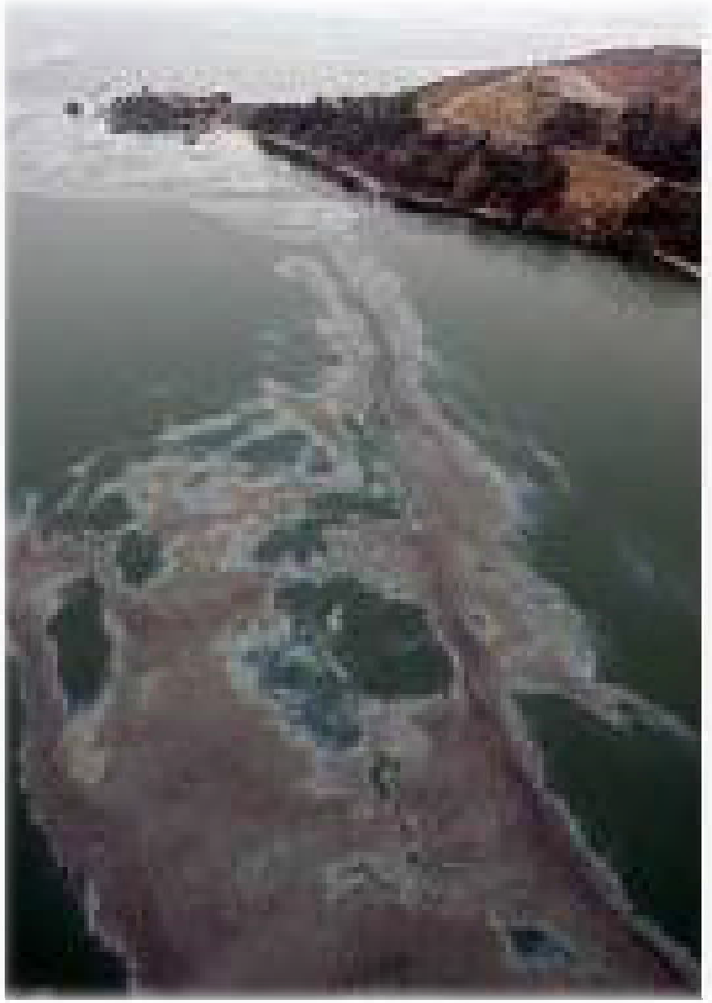
- Oil pollution from operational & accidental discharges;
- new oil tankers to have double hulls



Annex II

Regulations for the Control of Pollution by Noxious Liquid Substances in Bulk

- noxious liquid substances are carried in bulk;
- some 250 substances are evaluated and included
- discharge of the residues is allowed only to reception facilities
- no discharge of residues containing noxious substances is permitted within 12 miles of the nearest land.



Annex III

Prevention of Pollution by Harmful Substances Carried by Sea in Packaged Form

- requirements on packing, marking, labelling, documentation, stowage, quantity limitations, exceptions and notifications.
- “harmful substances” are those substances which are identified as marine pollutants in the International Maritime Dangerous Goods Code (IMDG Code) or which meet the criteria in the Appendix of Annex III.



Annex IV

Prevention of Pollution by Sewage from Ships

- requirements to control pollution of the sea by sewage;
- discharge of sewage into the sea is prohibited, except when the ship has in operation an approved sewage treatment plant at a more than 3 NM from the nearest land;
- sewage not disinfected has to be discharged at more than 12 NM from the nearest land.



Annex V

Prevention of Pollution by Garbage from Ships

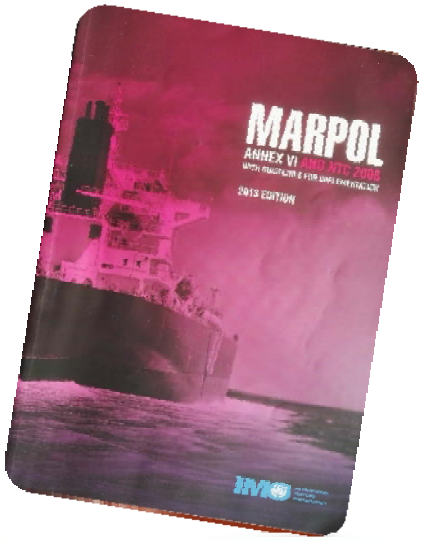
- disposal in different types of garbage and specifies
- distances from land and the manner
- complete ban imposed on the disposal into the sea of all forms of plastics.



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Annex VI

Prevention of Air Pollution from Ships



- set limits on sulphur oxide and nitrogen oxide emissions from ship exhausts
- prohibit deliberate emissions of ozone depleting substances;
- emission control areas set more stringent standards for SOx, NOx and particulate matter.
- mandatory technical and operational energy efficiency measures aimed at reducing greenhouse gas emissions from ships





STCW Convention with 2010 Manila Amendments

Adopted: 7 July 1978;
Entered into force: 28 April 1984;
Last revisions: 2010

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STCW Convention was the first to establish basic requirements on training, certification and watchkeeping for seafarers on an international level.

STCW prescribes minimum standards relating to training, certification and watchkeeping for seafarers which countries are obliged to meet or exceed.





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The Manila amendments to the STCW Convention and Code were adopted on 25 June 2010, marking a major revision of the STCW Convention and Code.

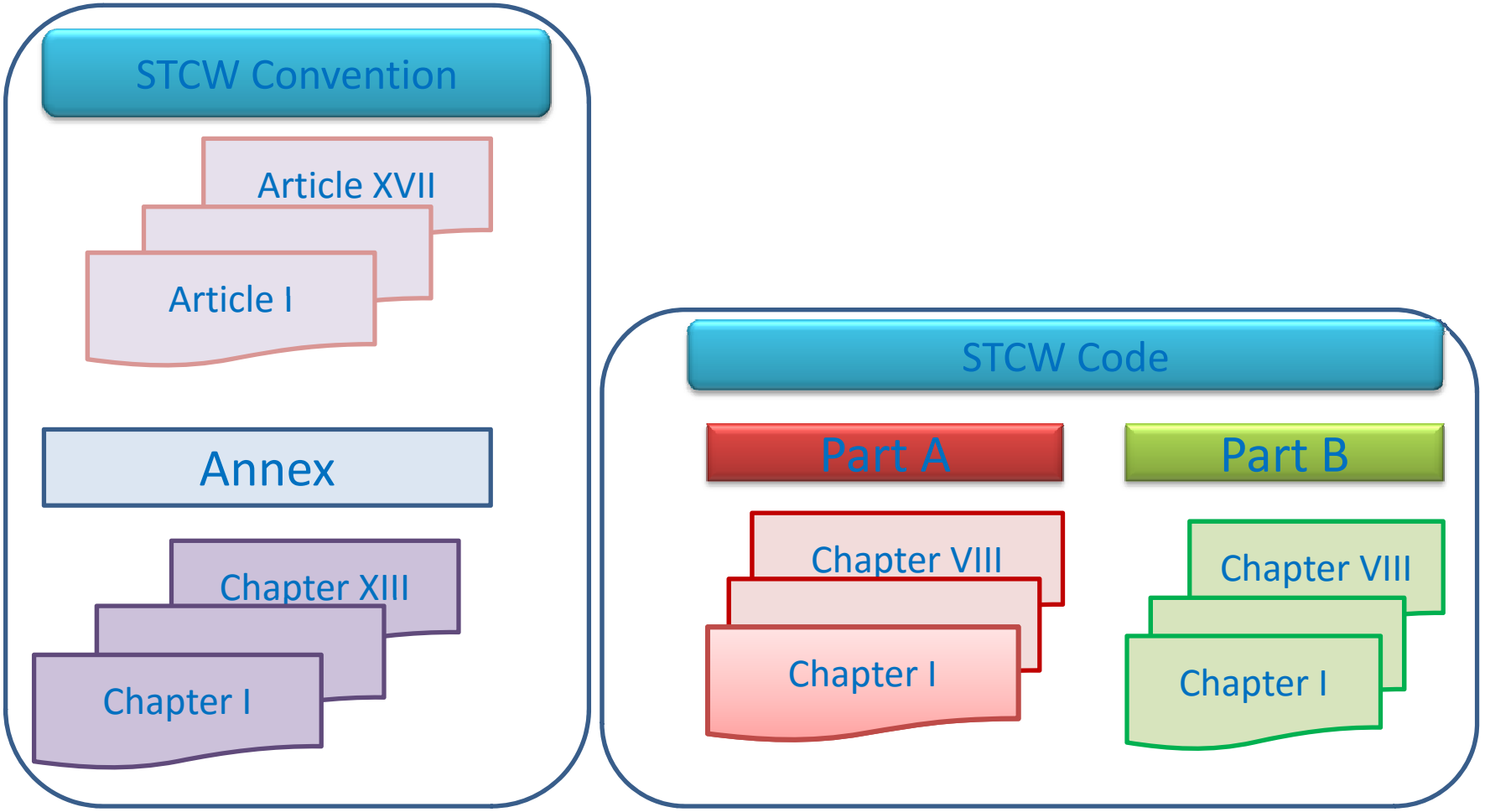
The 2010 amendments entered into force on 1 January 2012 under the tacit acceptance procedure, the amendments

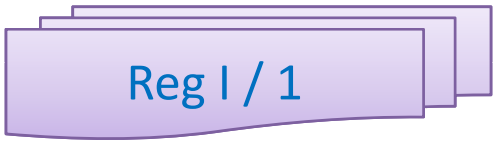
- are aimed at bringing the Convention and Code up to date with developments since they were initially adopted
- enable to address issues that are anticipated to emerge in the foreseeable future.

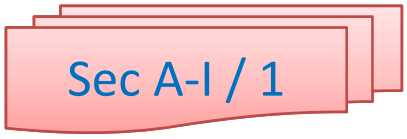
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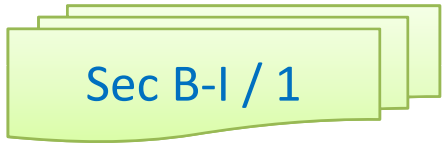




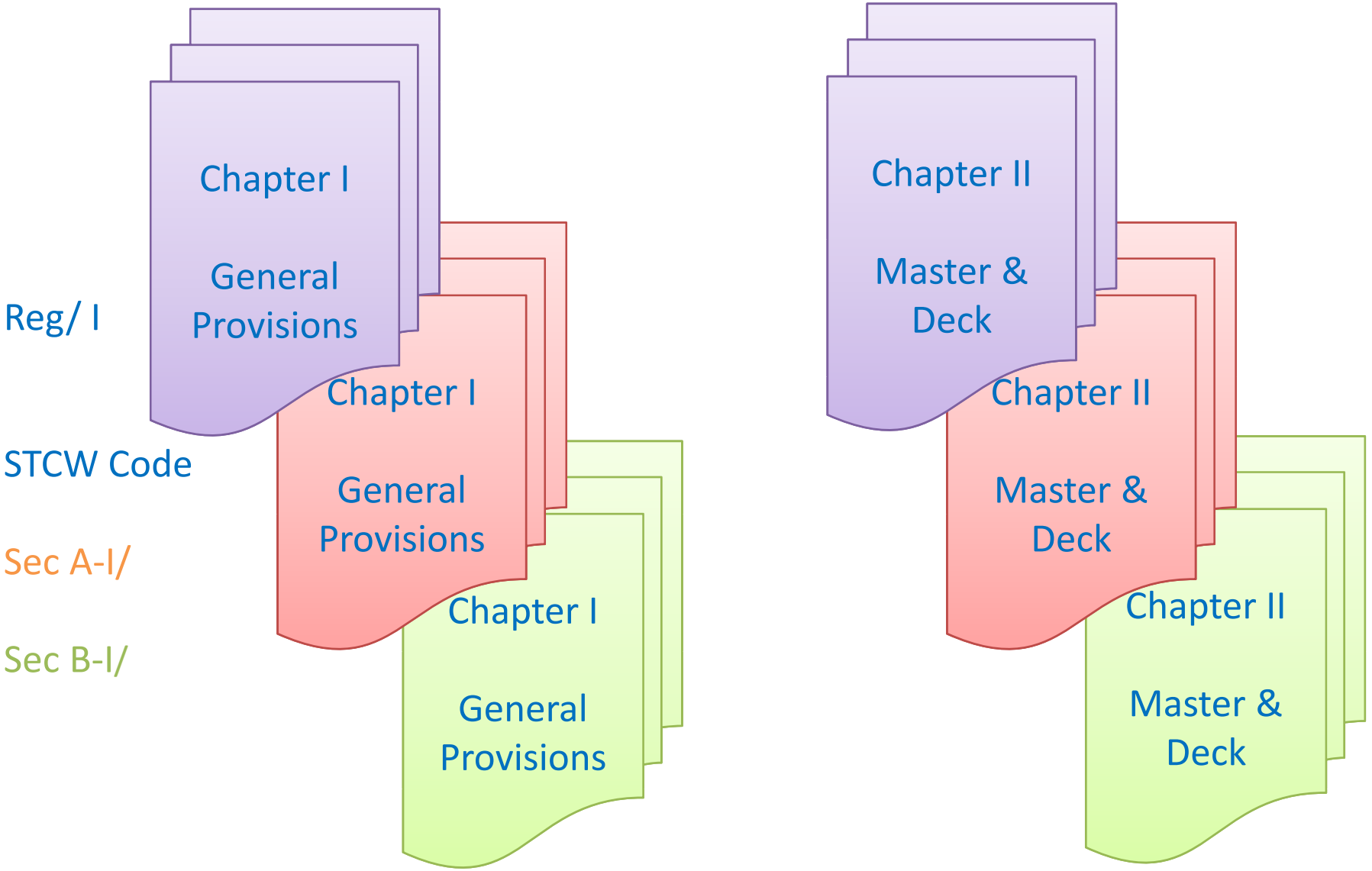




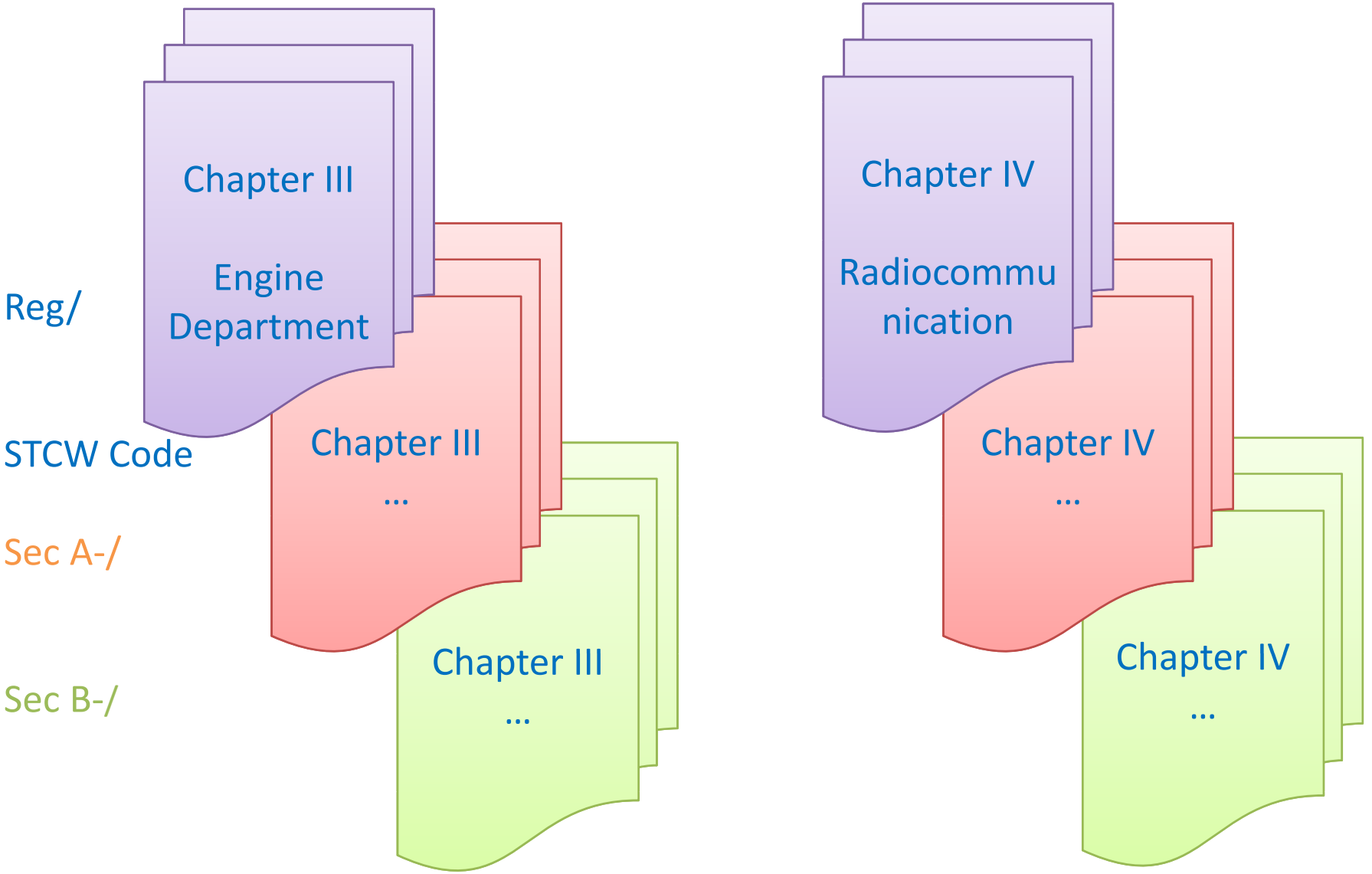




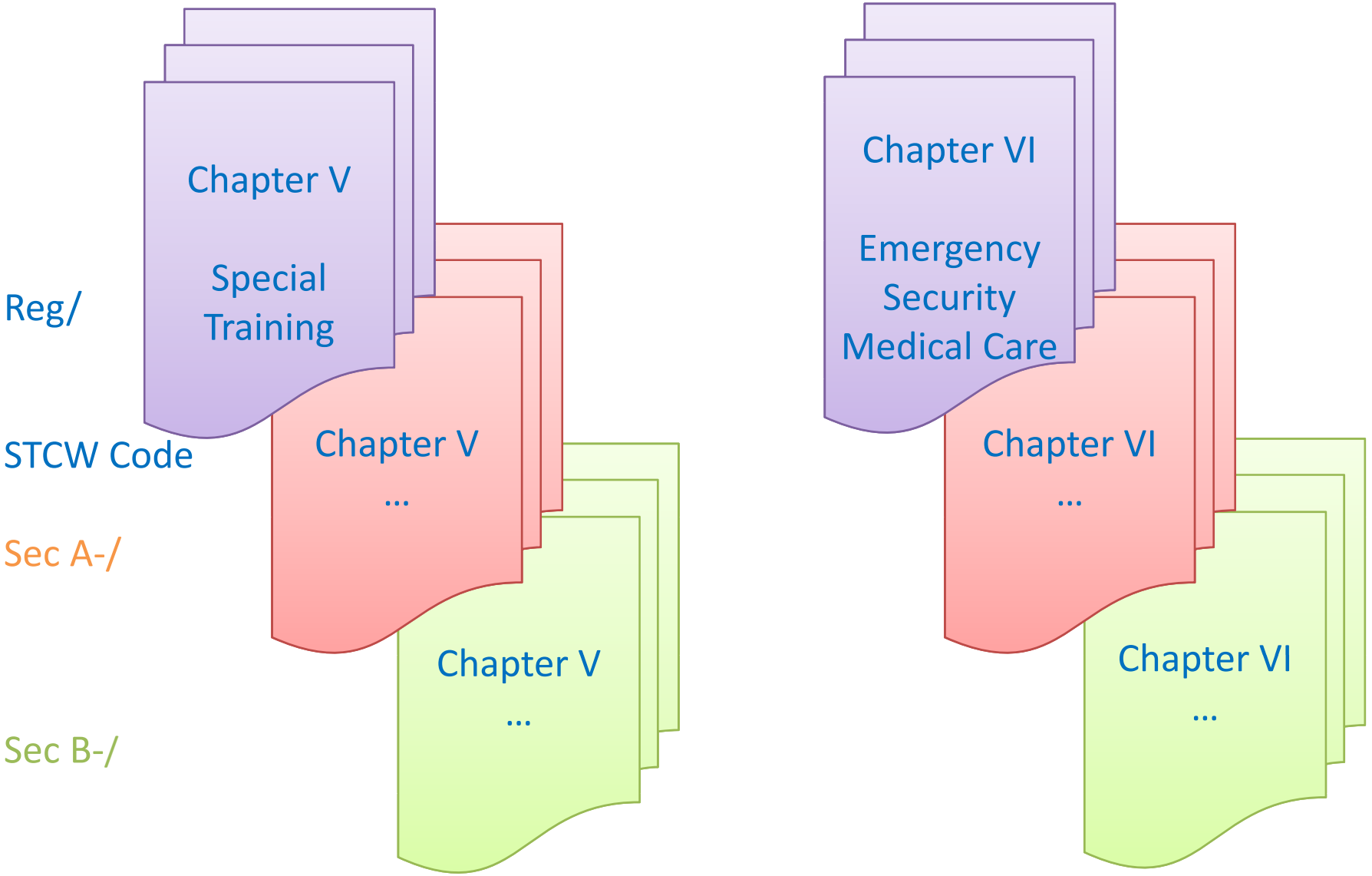
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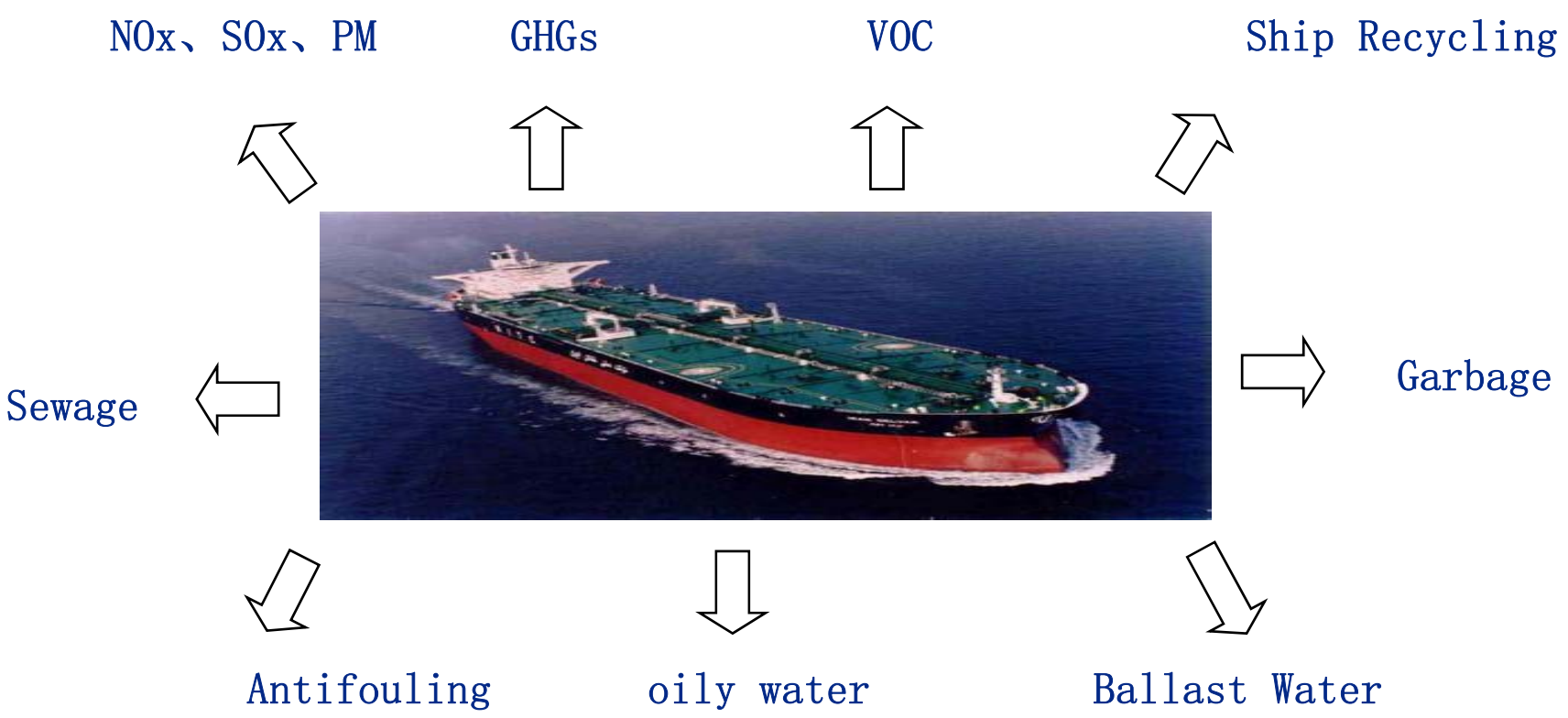


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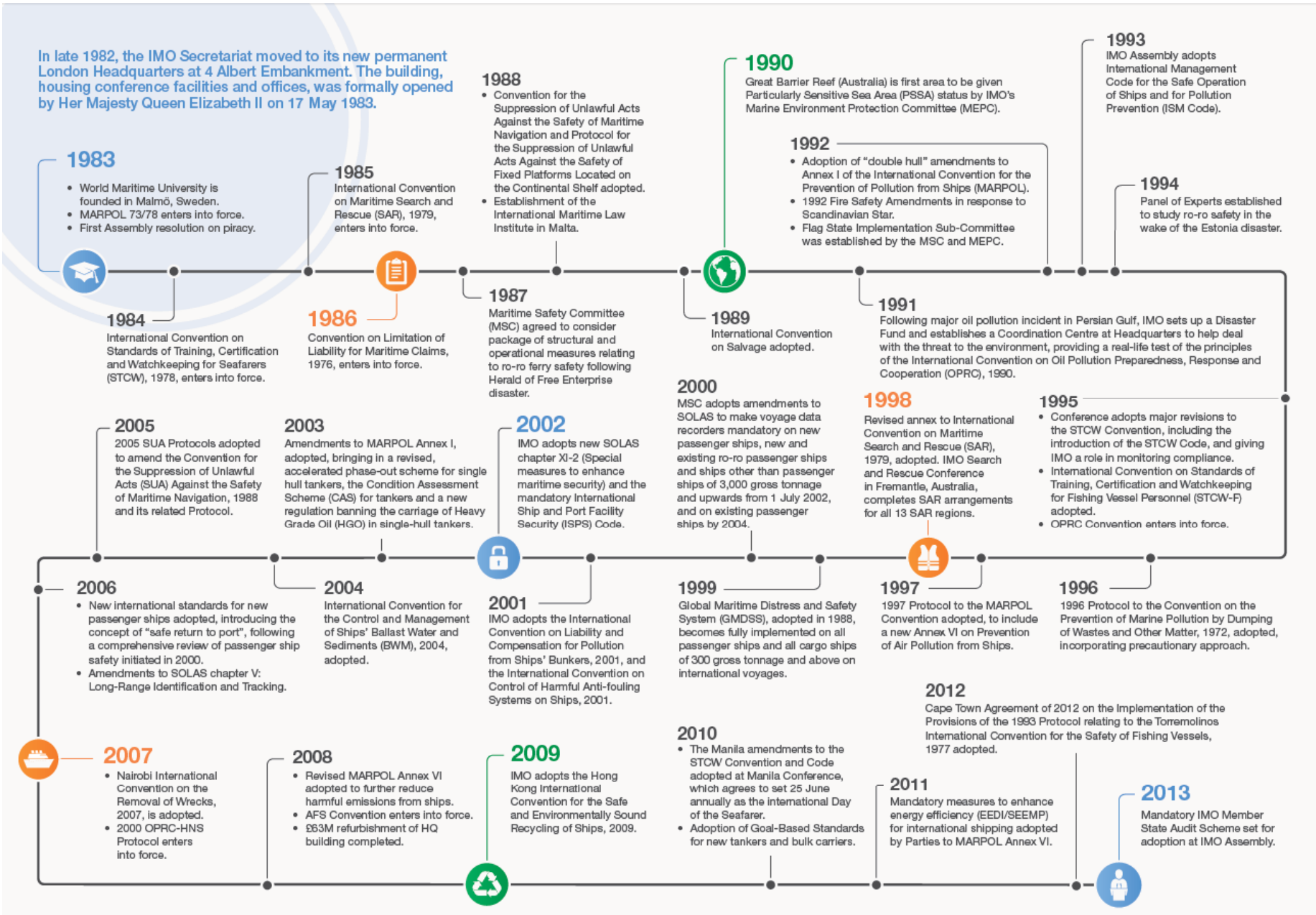




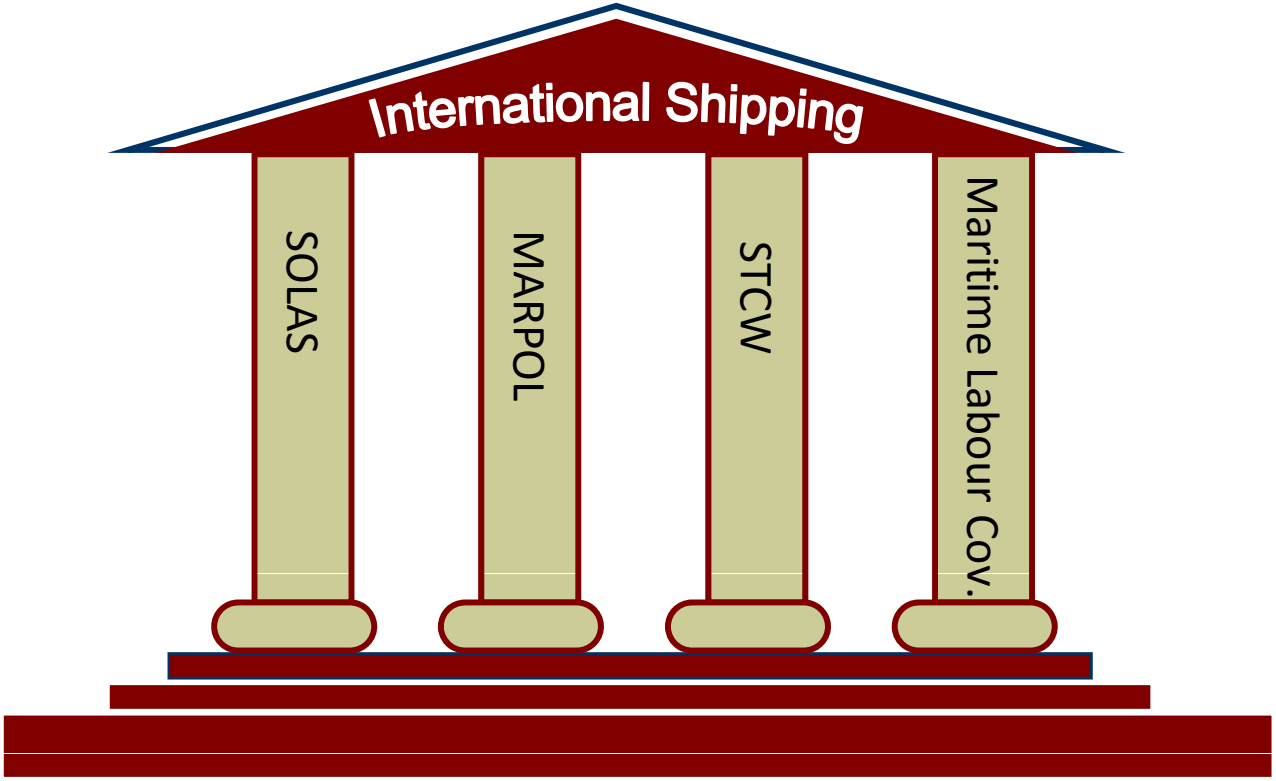
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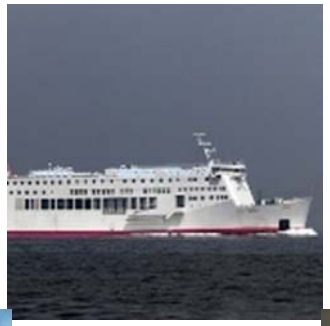


4. Current Hot Issues

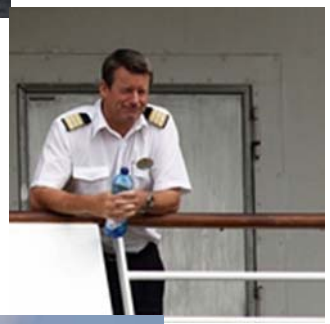


Maritime Safety & Security

- domestic ferry safety
- gas fuel safety – IGF Code
- fatigue guideline
- polar code training
- passenger ship safety
- GMDSS











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Marine Environment Protection

- Polar code – environment part
- fuel consumption data collection
- PSSA – Great Barrier Reef
- Ballast water roadmap
- BWM Convention
- fuel availability review
- oil spill response guidance

Legal Matters

- Wreck Removal Convention entered into force
- Limitation of liability for maritime claims
- Support for drafters and their national legislation







THANK YOU

Q & A